Digital Signal Processing Laboratory (EE39203)

Experiment 5:

Digital Filter Design

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3. (I)

> **Lab Report:**
> Submit the difference equation, system diagram, and the analytical expression of the impulse response for the filter $H_f(z)$. Also submit the plot of the magnitude response for the three values of θ. Explain how the value of θ affects the magnitude of the filter's frequency response.

a. Difference Equations:

$$H_f(z) = 1-(2\cos\theta)z^{-1} + z^{-2}$$

or, $$\frac{Y(z)}{X(z)} = 1 - 2\cos\theta \cdot z^{-1} + z^{-2}$$

[$Y(z)$ & $X(z)$ are the z-transforms of output & input]

or, $$Y(z) = X(z) - 2\cos\theta \cdot z^{-1} \cdot X(z) + z^{-2} \cdot X(z)$$

[We know that, $x[n] \xleftrightarrow{\text{z-transform}} X(z) = \sum_{n=-\infty}^{+\infty} z^{-n} \cdot x[n]$

$$x[n-n_0] \xleftrightarrow{\text{z-transform}} X_{n_0}(z) = \sum_{n=-\infty}^{+\infty} z^{-n} \cdot x[n-n_0]$$

$$= z^{-n_0} \sum_{n-n_0=-\infty}^{\infty} z^{-(n-n_0)} \cdot x[n-n_0]$$

$$= z^{-n_0} \cdot \sum_{k=-\infty}^{+\infty} z^{-k} \cdot x[k]$$

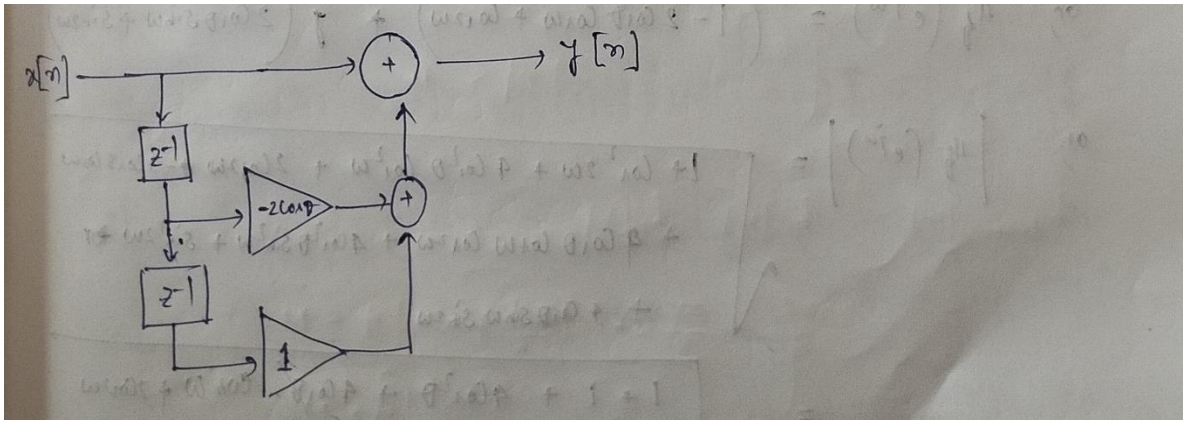
$$= z^{-n_0} \cdot X(z)$$]

Taking inverse z of the both sides we get -

or, $$\boxed{y[n] = x[n] - 2\cos\theta \cdot x[n-1] + x[n-2]}$$

difference equation for the system.

b. System Diagrams:



c. Analytical Expression of the Impulse Responses:

$$H_f(z) = 1 - 2\cos\theta \cdot z^{-1} + z^{-2}$$

We know that, $\delta[n] \xleftrightarrow{z} \sum_{n=-\infty}^{+\infty} \delta[n] z^{-n} = 1$

$\delta[n-n_0] \xleftrightarrow{z} \sum_{n=-\infty}^{+\infty} \delta[n-n_0] z^{-n} = z^{-n_0}$

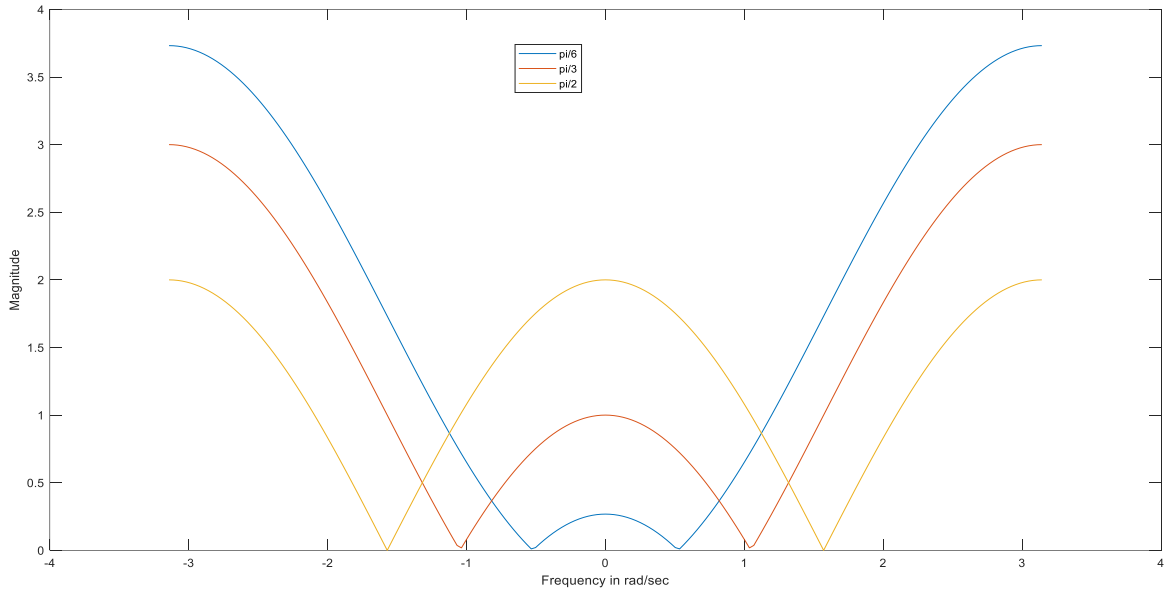
Taking inverse z-transform of both sides.

or, $h_f[n] = \delta[n] - 2\cos\theta \cdot \delta[n-1] + \delta[n-2]$

d. MATLAB Code and Plot of Magnitude Response for Different Thetas:

```
j=(-1)^0.5;
theta=[pi/6 pi/3 pi/2];
w=-pi:pi/100:pi;
for i=1:3
    DTFT_x=1-2.*cos(theta(i)).*exp(-j*w)+exp(-2*j*w);
    plot(w,abs(DTFT_x));
    ylabel('Magnitude');
    xlabel('Frequency in rad/sec');
    hold on;
```

```
end
```



e.

Explanation on how the value of θ affects the magnitude of the filter's frequency response:

$$H_f(z) = 1 - 2\cos\theta\, z^{-1} + z^{-2}$$

or) $$H_f(e^{j\omega}) = 1 - 2\cos\theta\, e^{-j\omega} + e^{-j2\omega}$$

or) $$H_f(e^{j\omega}) = 1 - 2\cos\theta\,(\cos\omega - j\sin\omega) + (\cos 2\omega - j\sin 2\omega)$$

or) $$H_f(e^{j\omega}) = (1 - 2\cos\theta\cos\omega + \cos 2\omega) + j(2\cos\theta\sin\omega - \sin 2\omega)$$

or) $$|H_f(e^{j\omega})| = \sqrt{1 + 4\cos^2\theta\cos^2\omega + \cos^2 2\omega - 4\cos\theta\cos\omega - 4\cos\theta\cos\omega\cos 2\omega + 2\cos^2 2\omega + 4\cos^2\theta\sin^2\omega + \sin^2 2\omega - 4\cos\theta\sin\omega\sin 2\omega}$$

$$= \sqrt{1 + 4\cos^2\theta + 1 - 4\cos\theta\cos\omega - 4\cos\theta\cdot\cos(2\omega - \omega) + 2\cos 2\omega}$$

$$= \sqrt{2 + 4\cos^2\theta + 2\cos 2\omega - 8\cos\theta\cos\omega}$$

$$= \sqrt{2 + 4\cos\theta(\cos\theta - 2\cos\omega) + 2\cos 2\omega}$$

If, $\cos\theta - 2\cos\omega > 0$, or, $\omega > \cos^{-1}\left(\frac{1}{2}\cos\theta\right)$

then, $\theta\uparrow \Rightarrow \cos\theta\downarrow \Rightarrow \cos\theta\cdot(\cos\theta - 2\cos\omega)\downarrow \Rightarrow |H_f(e^{j\omega})|\downarrow$

If $\cos\theta - 2\cos\omega < 0$, or, $\omega < \cos^{-1}\left(\frac{1}{2}\cos\theta\right)$

then, $\theta\uparrow \Rightarrow \cos\theta\downarrow \Rightarrow \cos\theta\cdot(\cos\theta - 2\cos\omega)\uparrow \Rightarrow |H_f(e^{j\omega})|\uparrow$

So, we can see that if $\theta\uparrow$ then in a certain range (lower frequency) the magnitude increases and in outer range magnitude decreases.

3.(II)

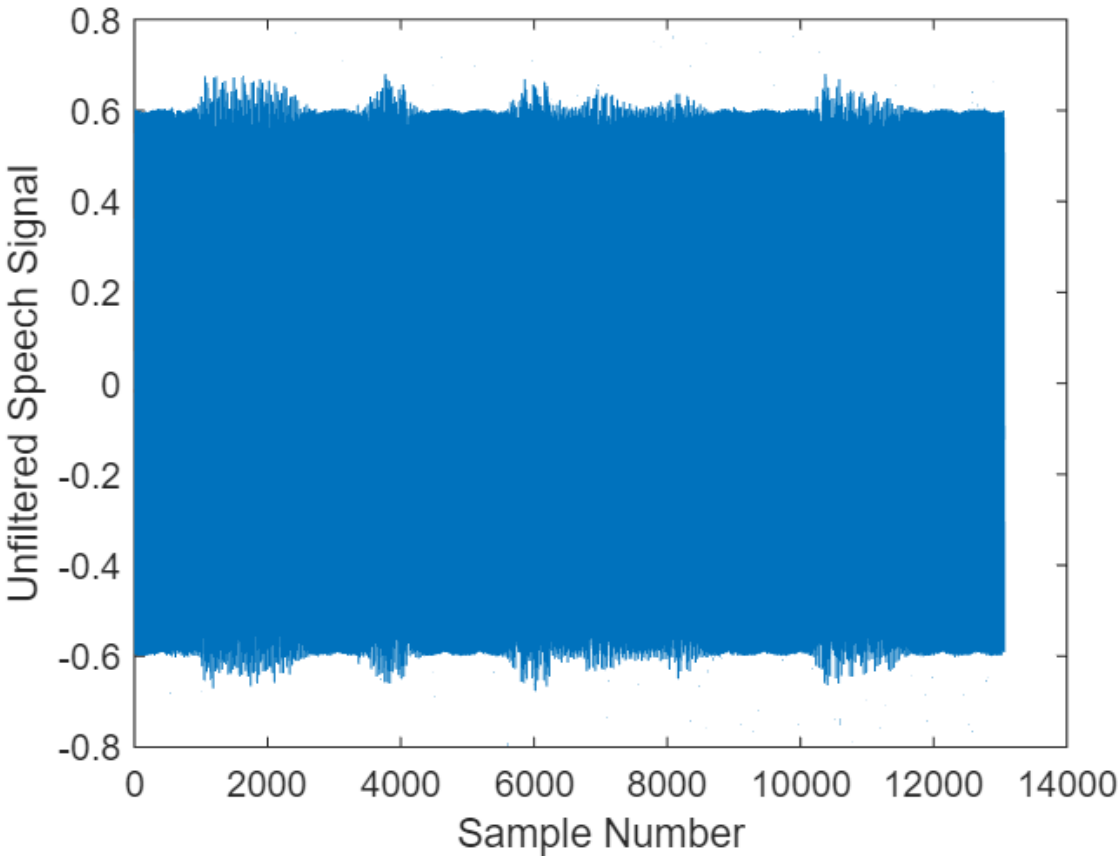
Lab Report:
For both the original audio signal and the filtered output, hand in the following:

- The time domain plot of 101 samples.
- The plot of the magnitude of the DTFT for 1001 samples.

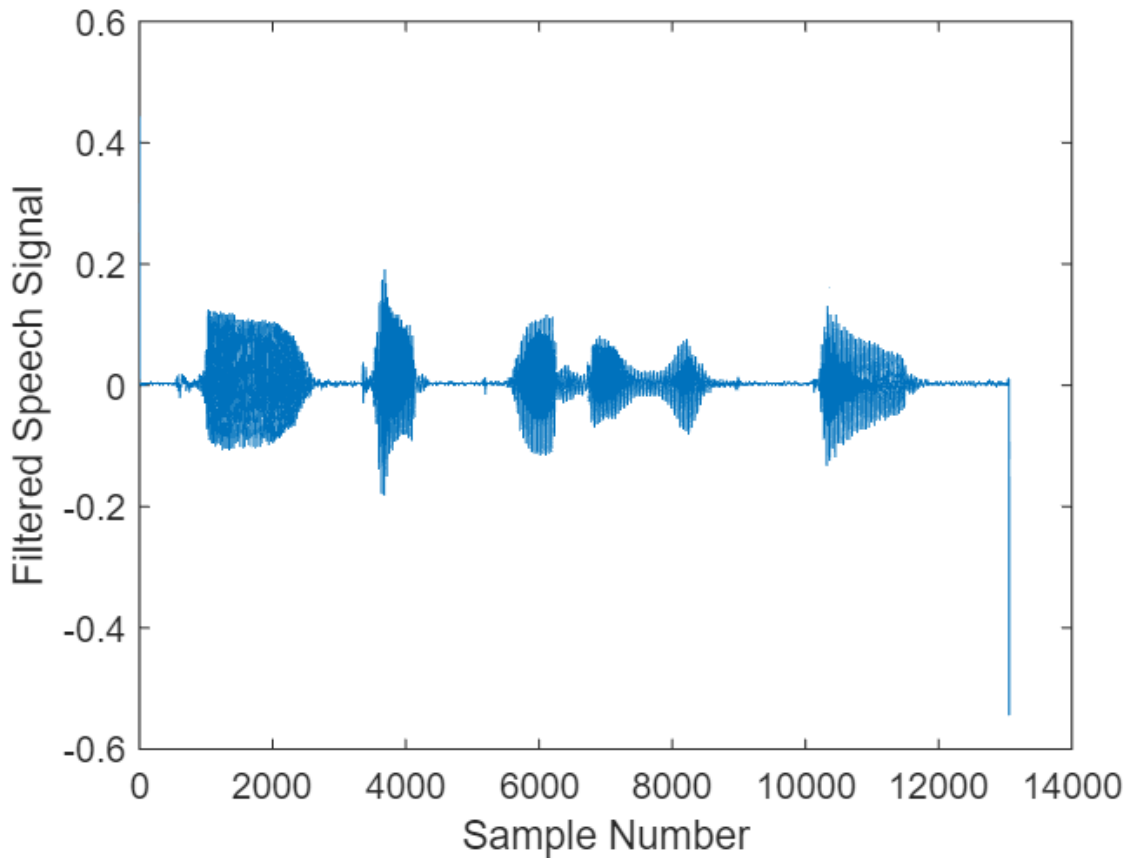
Also hand in the code for the FIRfilter filtering function. Comment on how the frequency content of the signal changed after filtering. Is the filter we used a lowpass, highpass, bandpass, or a bandstop filter? Comment on how the filtering changed the quality of the audio signal.

a. Time domain Plot of samples:

i. When filtering is not there:

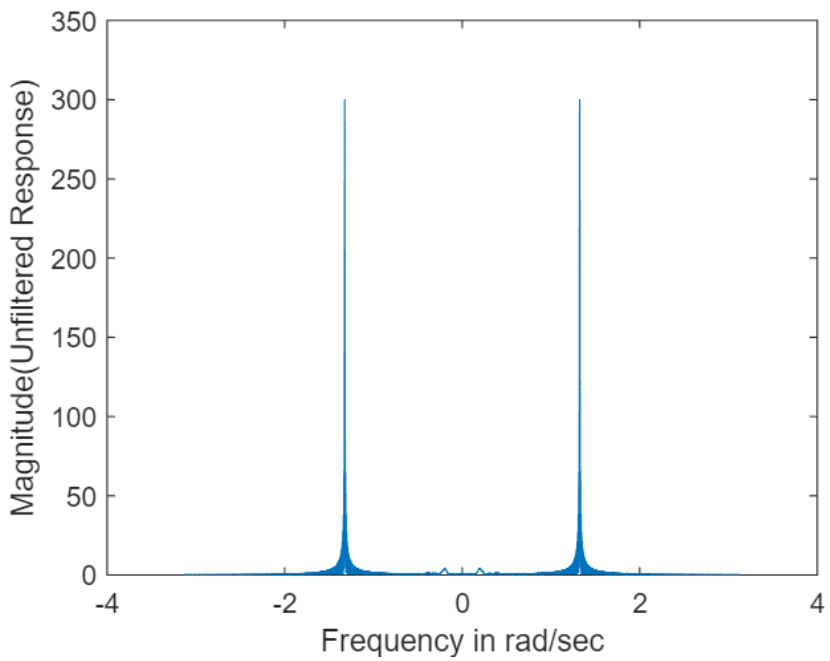


ii. When filtering has been done:

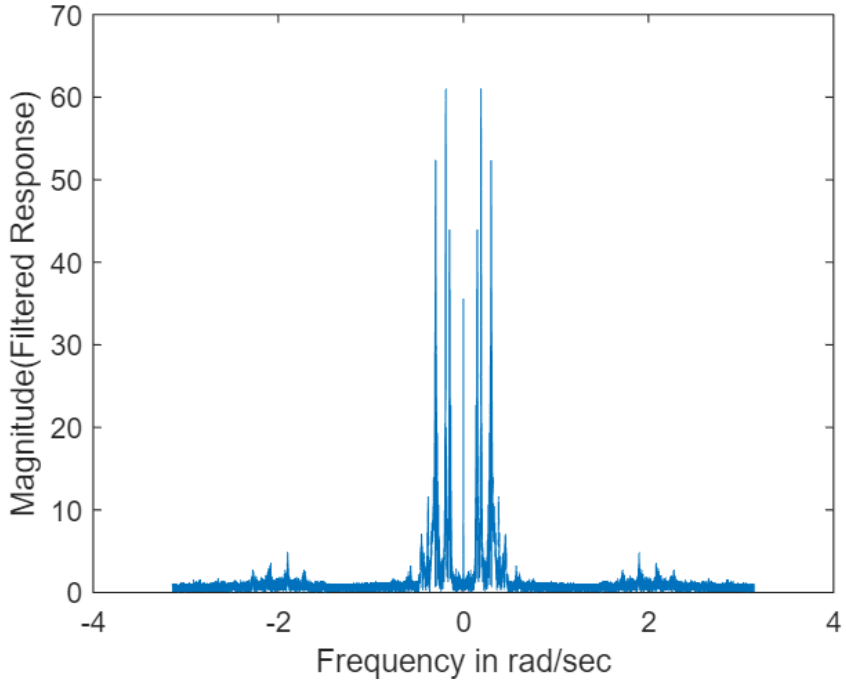


b. Plot of Magnitude of DTFT for 1001 Samples:

i. When there is no filtering:



ii. When filtering has been done:



c. MATLAB Code for performing the upper two tasks:

```
y=load("C:\Users\PANTHADIP MAJI\Desktop\DSP_LAB\Exp_5\nspeech1.mat");
x1=zeros(length(y.nspeech1),1);
x2=zeros(length(y.nspeech1),1);
plot(y.nspeech1);
ylabel('Unfiltered Speech Signal');
xlabel('Sample Number');


for i=1:length(y.nspeech1)%Storing only 100th to 1100th samples
    if i>=100 && i<=1100
        x1(i)=y.nspeech1(i);
    else
        x1(i)=0;
    end
end
[x1_DTFT,W]=DTFT(x1,0);%Applying DTFT to the stored 1001 samples
plot(W,abs(x1_DTFT));%Plotting DTFT of the 1001 samples
ylabel('Magnitude(Unfiltered Response)');
xlabel('Frequency in rad/sec');


%Calculation and display of the dominant frequencies
[x1_max,pos_max]=max(abs(x1_DTFT));
display(x1_max);
display(pos_max);
```

```
w=W(pos_max);
display(w);


%Applying FIR filter function and plotting the filtered signal
x2=FIRfilter(y.nspeech1);
plot(x2);
ylabel('Filtered Speech Signal');
xlabel('Sample Number');
sound(x2);
[x2_DTFT,W1]=DTFT(x2,0);
plot(W1,abs(x2_DTFT));%Plot of the DTFT of the filtered signal
ylabel('Magnitude(Filtered Response)');
xlabel('Frequency in rad/sec');
```

Here two functions are being used, i.e., FIRfilter & DTFT.

```
function x_modified=FIRfilter(x)% Code for FIRfilter function
N=length(x);
h=[1 -0.48976 1];
x1=zeros(N,1);
for i=0:2% Convolution of the impulse response h and the input x
    for j=1:N
        if j+i<=0 || j+i>N
            break;
        elseif j<i
            x1(j)=x1(j)+0;
        else
            x1(j)=x1(j)+(h(i+1)*x(j+i));
        end
    end
end
x_modified=x1;
end

function [X,w] = DTFT(x,M)%Code for DTFT function
% This function computes samples of the DTFT of x.
% To compute the DTFT of x, use
%
%             [X,w] = DTFT(x,0)
%
% where X is the vector of DTFT samples and w is the
% vector of radial frequencies. To compute at least
% M samples of the DTFT, you may use the command
%
%             [X,w] = DTFT(x,M)
%
% This is useful when the plot of X versus w does
% not contain a sufficient number of points.


N = max(M,length(x));
N = 2^(ceil(log(N)/log(2)));
```

```
% Take the padded fft
X = fft(x,N);
w = 2*pi*( (0:(N-1))/N );
w = w - 2*pi*(w>=pi);


% Shift FFT to go from -pi to pi
X = fftshift(X);
w = fftshift(w);
end
```

d. What type of filter it was?

Ans: This is a notch type of filter which deletes only the required frequency component from the signal. But we cannot say it as a pure notch filter because other than the deleting frequency the magnitude should be the same over the whole range of frequency but in this case, it is not.

e. Comment:

After filtering the noise frequency got removed by the filter and hence, we separate out the actual signal. We can see also the difference in the time domain plot of unfiltered and filtered waveforms.

f. The Dominant Frequency(w) in the unfiltered signal:

```
x1_max = 300.2636
pos_max = 4742
w = -1.3234
```

4.(I)

Lab Report:
Submit the difference equation, system diagram and the analytical expression of the impulse response for $H_i(z)$. (Hint: The frequency response of the system can be obtained by restricting the z-transform to the unit circle. So the DTFT of $h_i[n]$ is $H_i(e^{j\omega})$. Therefore, to get $h_i[n]$, you can take the inverse Fourier transform of $H_i(e^{j\omega})$.) Also submit the plot of the magnitude of the frequency response for each value of r. Explain how the value of r affects this magnitude.

a. Difference Equation:

(i) $$H_i(z) = \frac{1-r}{1-2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$

or, $$\frac{Y(z)}{X(z)} = \frac{1-r}{1-2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$

[$Y(z)$ & $X(z)$ are the z-transforms of ~~input~~ output & input]

or, $$Y(z)\left\{1-(2r\cos\theta)z^{-1} + r^2 z^{-2}\right\} = X(z)(1-r)$$

or, $$Y(z) - (2r\cos\theta)\cdot z^{-1}\cdot Y(z) + r^2 z^{-2}\cdot Y(z) = (1-r)X(z)$$

[Now we know that,

$$x[n] \xleftrightarrow{\text{z-Transform}} X(z) = \sum_{n=-\infty}^{+\infty} z^{-n}\cdot x[n]$$

$$x[n-n_0] \xleftrightarrow{\text{z-Transform}} X_{n_0}(z) = \sum_{n=-\infty}^{+\infty} z^{-n}\cdot x[n-n_0]$$

[Let's say $n-n_0 = k$]

$$= z^{-n_0}\cdot \sum_{n-n_0=-\infty}^{+\infty} z^{-(n-n_0)}\cdot x[n-n_0]$$

$$= z^{-n_0}\cdot \sum_{k=-\infty}^{+\infty} z^{-k}\cdot x[k]$$

$$= z^{-n_0}\cdot X(z)$$]

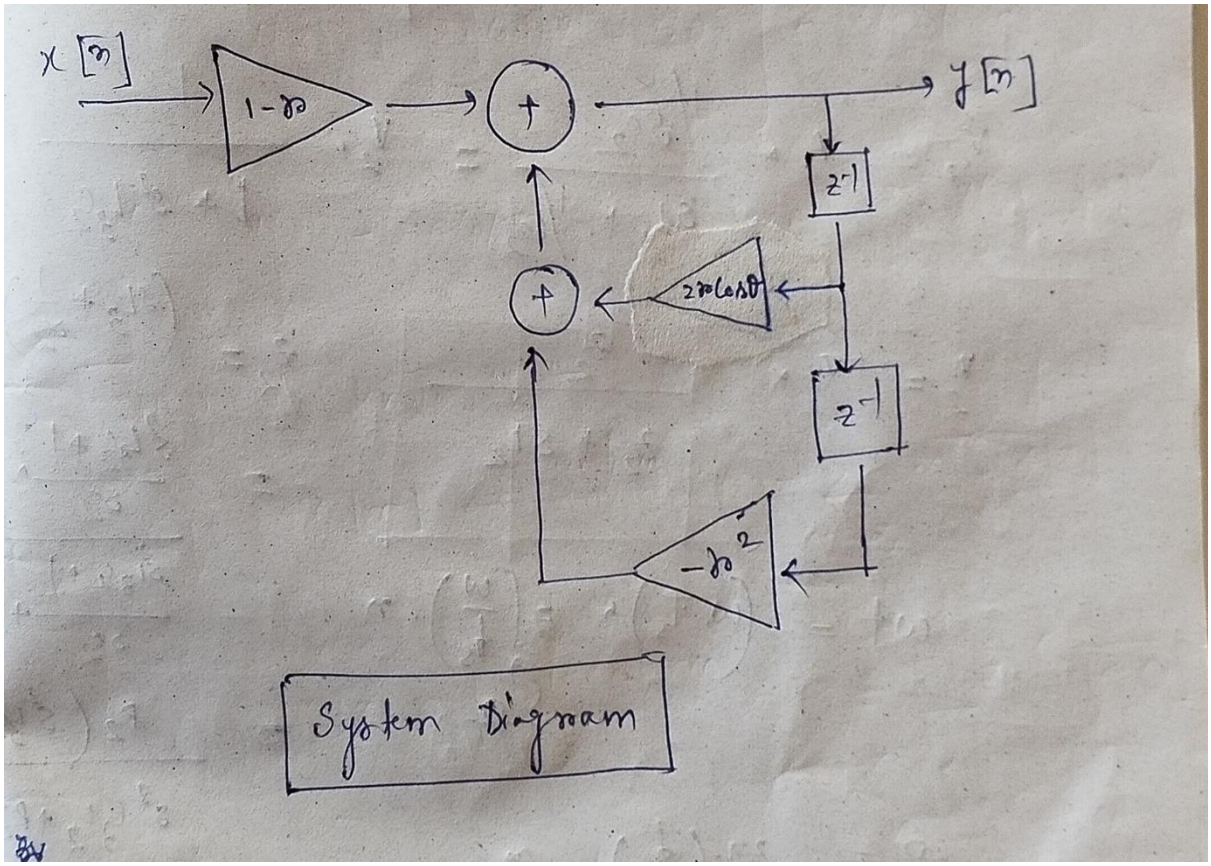
Taking inverse z of the both sides we get -

or, $$y[n] - 2r\cos\theta\, y[n-1] + r^2 y[n-2] = (1-r)x[n]$$

or, $$\boxed{y[n] = 2r\cos\theta\cdot y[n-1] - r^2\cdot y[n-2] + (1-r)\cdot x[n]}$$

Difference Equation for the system

b. System Diagram:



c. Analytical expression for the impulse response of $H_i(z)$:

(ii) $$H_i(z) = \frac{1-r}{\left(1-re^{j\theta}z^{-1}\right)\left(1-re^{-j\theta}z^{-1}\right)}$$

$$= \frac{1-r}{1-e^{2j\theta}} \cdot \frac{\left(1-re^{j\theta}z^{-1}\right) - e^{2j\theta}\left(1-re^{-j\theta}z^{-1}\right)}{\left(1-re^{j\theta}z^{-1}\right)\left(1-re^{-j\theta}z^{-1}\right)}$$

$$= \frac{1-r}{1-e^{2j\theta}} \left[\frac{1}{\left(1-re^{-j\theta}z^{-1}\right)} - \frac{e^{2j\theta}}{\left(1-re^{j\theta}z^{-1}\right)}\right]$$

Taking inverse Z-Transform we get,

$$h_i[n] = \frac{1-r}{1-e^{2j\theta}} \cdot \left[\left(re^{-j\theta}\right)^n u[n] - e^{2j\theta} \cdot \left(re^{j\theta}\right)^n u[n]\right]$$

or, $$h_i[n] = \frac{1-r}{1-e^{2j\theta}} \cdot r^n \cdot u[n] \cdot \left[e^{-j\theta n} - e^{2j\theta} \cdot e^{j\theta n}\right]$$

$$= \frac{(1-r) \cdot r^n}{1-e^{2j\theta}} \cdot e^{j\theta} \cdot \left(e^{-j\theta(n+1)} - e^{j\theta(n+1)}\right) u[n]$$

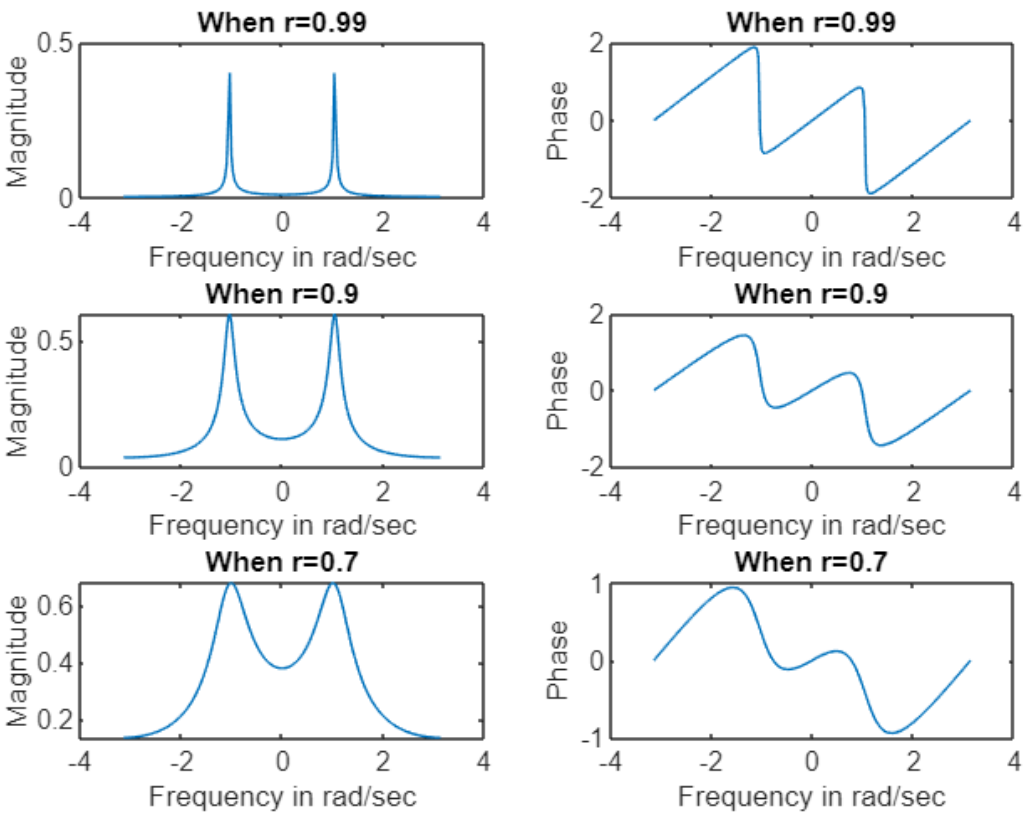
$$= \frac{(1-r) \cdot r^n}{e^{-j\theta} - e^{j\theta}} \cdot \left\{e^{-j\theta(n+1)} - e^{j\theta(n+1)}\right\} u[n]$$

$$= \frac{(1-r)\, r^n}{\left(\frac{e^{-j\theta} - e^{j\theta}}{-2j}\right)} \cdot \left(\frac{e^{-j\theta(n+1)} - e^{j\theta(n+1)}}{-2j}\right) u[n]$$

$$= \frac{(1-r)\cdot r^n}{\sin\theta} \cdot \sin\{\theta(n+1)\} \quad u[n]$$

$$= \frac{(1-r)\, r^n \cdot \sin\{\theta(n+1)\}}{\sin\theta} \cdot u[n]$$

d. Plot of magnitude response for different values of r:




```
j=(-1)^0.5;
theta=pi/3;
r=[0.99 0.9 0.7];
w=-pi:pi/100:pi;
for i=1:3
    DTFT_x=(1-r(i))./(1-2*r(i)*cos(theta)*exp(-j*w)+((r(i))^2)*exp(-2*j*w));
    subplot(3,2,2*(i-1)+1);
```

```
    plot(w,abs(DTFT_x));
    xlabel('Frequency in rad/sec');
    ylabel('Magnitude');
    if i==1
    title('When r=0.99')
    else if i==2
    title('When r=0.9')
    else
    title('When r=0.7')
    end
    end
    subplot(3,2,2*i);
    plot(w,angle(DTFT_x));
    xlabel('Frequency in rad/sec');
    ylabel('Phase');
    if i==1
    title('When r=0.99')
    else if i==2
    title('When r=0.9')
    else
    title('When r=0.7')
    end
    end
end
```

e. Explanation:

We saw that if r increases then the overall magnitude response decreases and the response at a certain frequency is becoming sharp and for other frequencies, it is lowered.

4. (II)

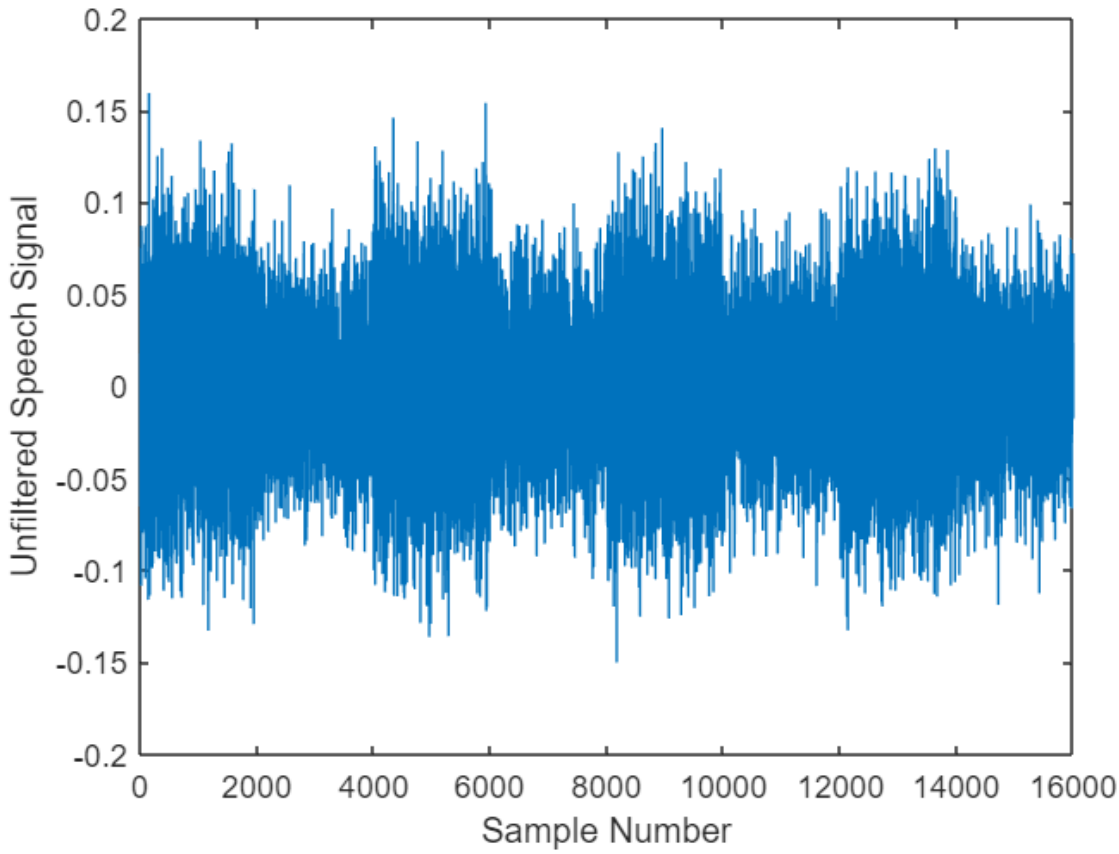
Lab Report:
For both the pcm signal and the filtered output, submit the following:

- The time domain plot of the signal for 101 points.
- The plot of the magnitude of the DTFT computed from 1001 samples of the signal.
- The plot of the magnitude of the DTFT for ω in the range $[\theta - 0.02, \theta + 0.02]$.

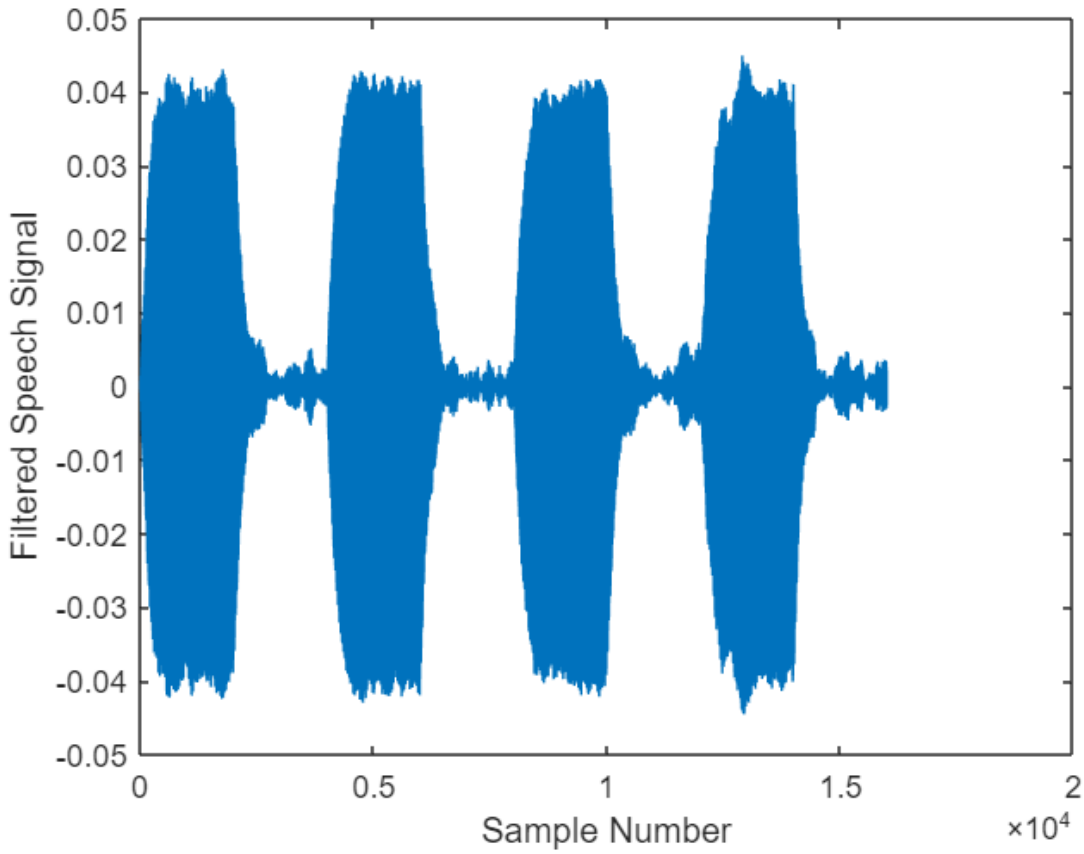
Also hand in the code for the `IIRfilter` filtering function. Comment on how the signal looks and sounds before and after filtering. How would you expect changes in r to change the filtered output? Would a value of r = 0.9999999 be effective for this application? Why might such a value for r be ill-advised? (Consider the spectrum of the desired signal around $\omega = \theta$.)

a. Time domain Plot:

i. When signal is not filtered:

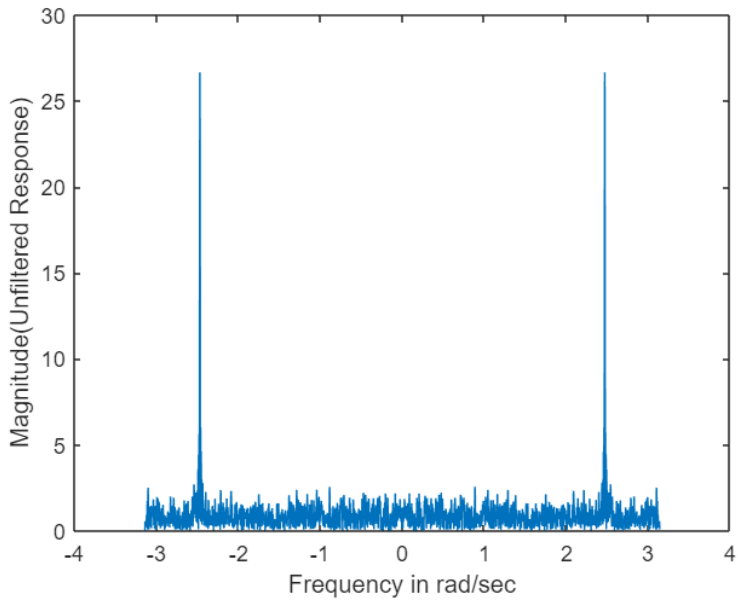


ii. When signal is filtered:

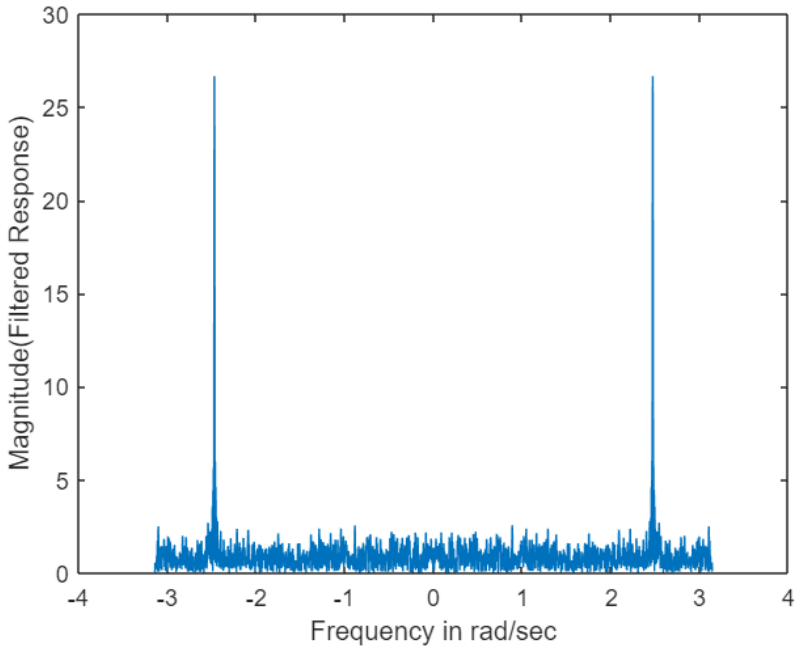


b. Plot of DTFT of 1001 samples:

i. When signal is not filtered:

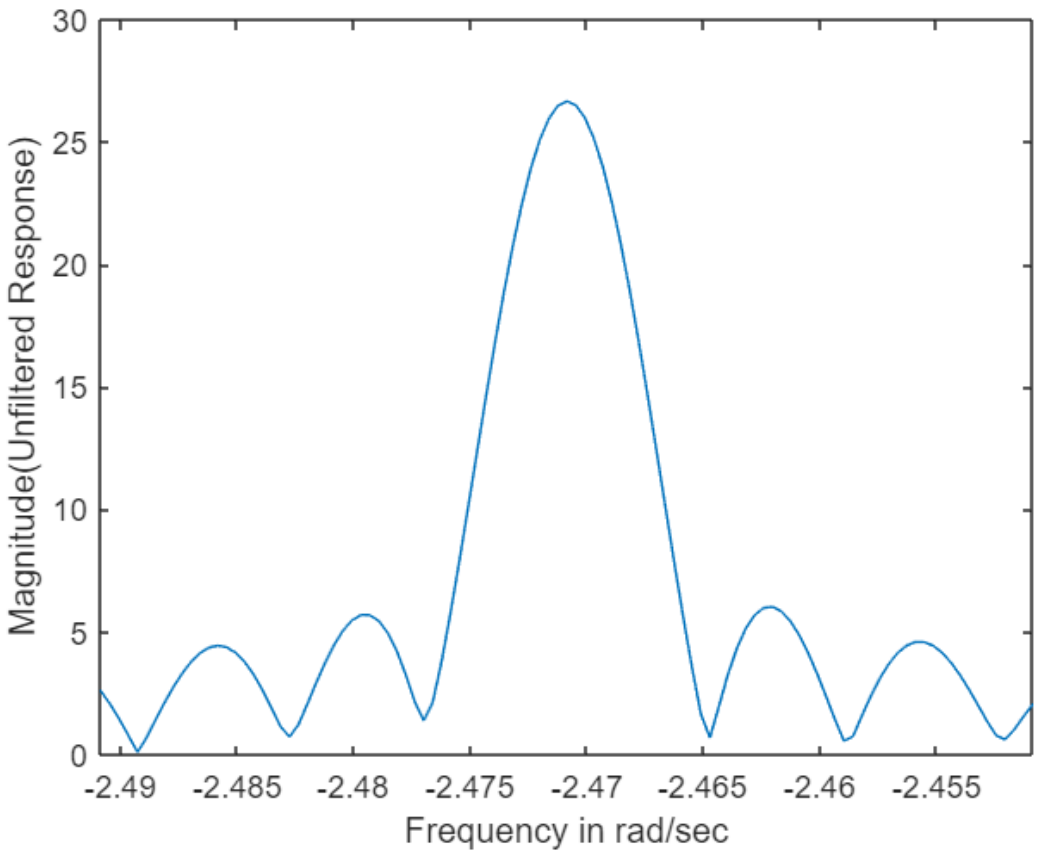


ii. When signal is filtered:

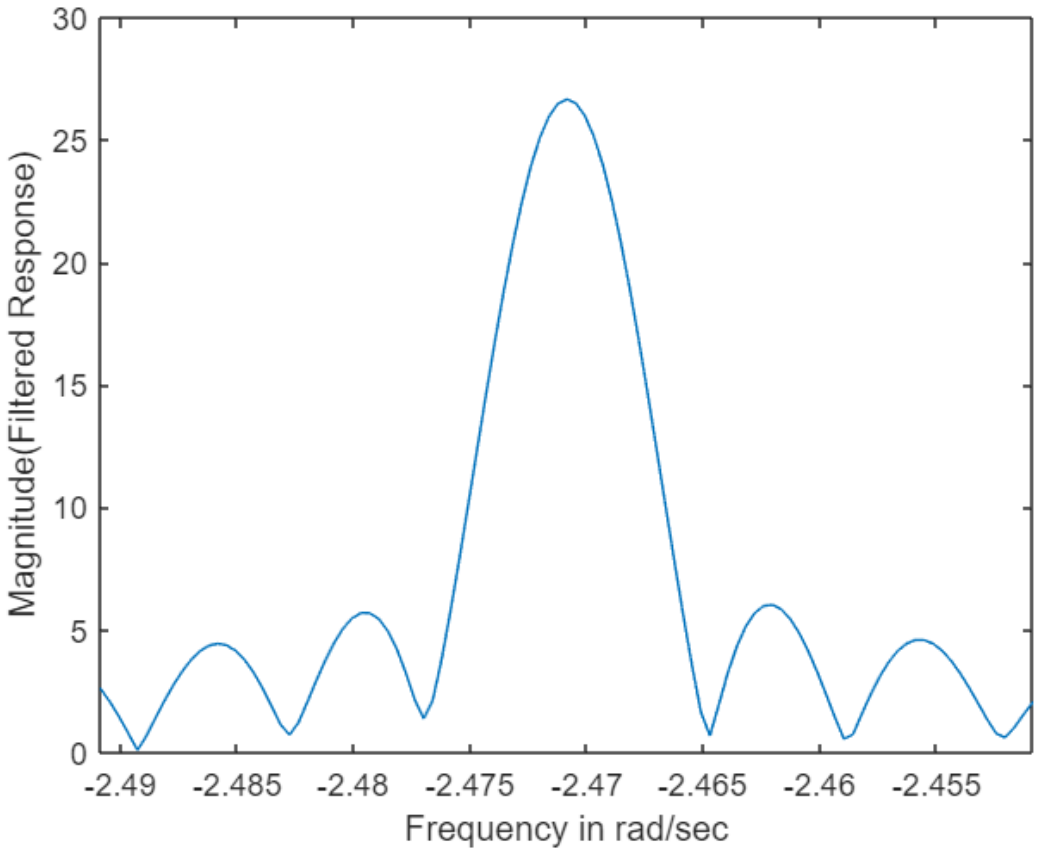


c. Plot of DTFT in the range of $\theta \pm 0.02$:

i. When signal is not filtered:



ii. When signal is filtered:



d. Filtering Function:

```
function x_modified=IIRfilter(x,theta,r)
%This function filters a signal 'x' with IIR filter of parameter theta & r
N=length(x)
y=zeros(N+2,1);
for i=1:N
y(i+2)=(1-r)*x(i)+2*r*cos(theta)*y(i+1)-(r^2)*y(i);%Recursive way of filtering
end
x_modified=y;
end
```

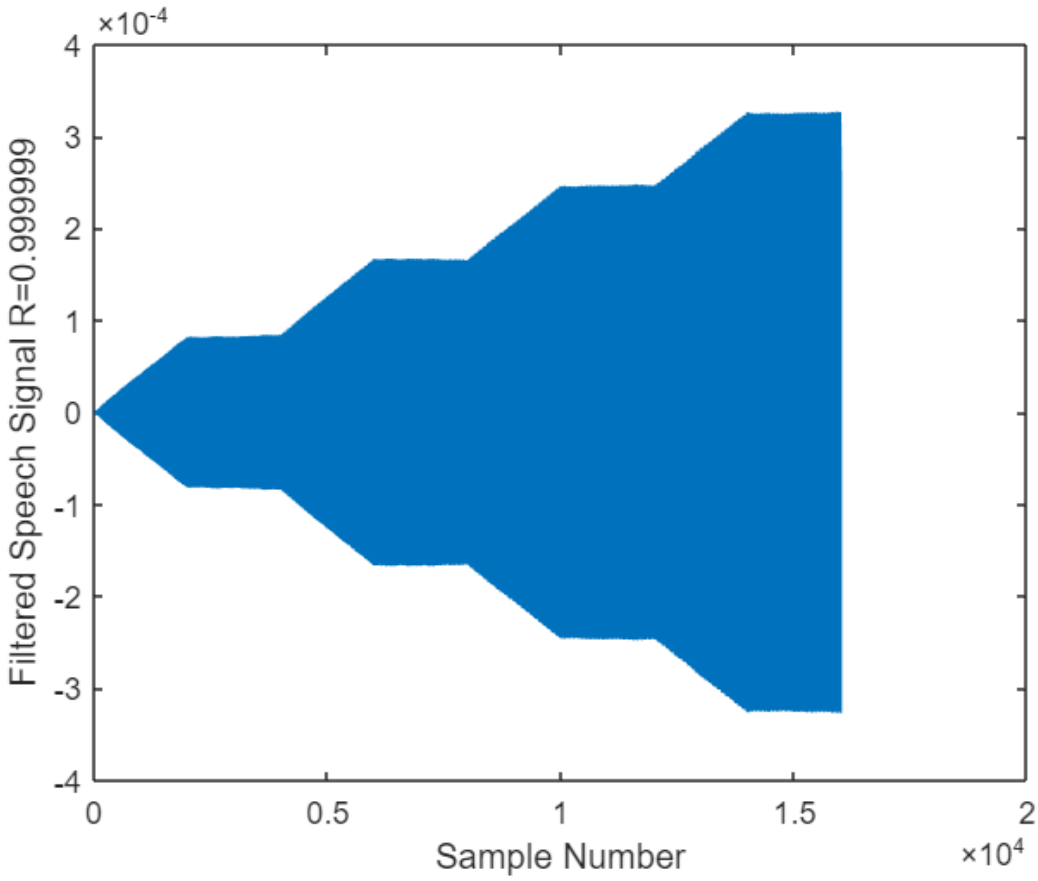
*MATLAB Code for performing the upper task:

```
y=load("C:\Users\PANTHADIP MAJI\Desktop\DSP_LAB\Exp_5\pcm.mat");
x1=zeros(length(y.pcm),1);
x2=zeros(length(y.pcm),1);
plot(y.pcm);%Plotting the unfiltered speech signal
ylabel('Unfiltered Speech Signal');
xlabel('Sample Number');
sound(y.pcm);
```

```
for i=1:length(y.pcm)%Storing 1001 sample values
    if i>=100 && i<=1100
        x1(i)=y.pcm(i);
    else
        x1(i)=0;
    end
end


[x1_DTFT,W]=DTFT(x1,0);
plot(W,abs(x1_DTFT));%Plotting DTFT of the unfiltered signal
ylabel('Magnitude(Unfiltered Response)');
xlabel('Frequency in rad/sec');


%Finding and displaying the dominant frequencies
[x1_max,pos_max]=max(abs(x1_DTFT));
display(x1_max);
display(pos_max);
w=W(pos_max);%w will store the dominant frequency
display(w);
plot(W,abs(x1_DTFT));%Plotting DTFT of the unfiltered signal in a cetain range
around w
xlim([w-0.02,w+0.02]);
ylabel('Magnitude(Unfiltered Response)');
xlabel('Frequency in rad/sec');


y=load("C:\Users\PANTHADIP MAJI\Desktop\DSP_LAB\Exp_5\pcm.mat");
x=IIRfilter(y.pcm,w,0.995);%The first term in the function is the signal to be
filtered and the others are the filter qualitative parameters
sound(x);
plot(x);%Plot of the filtered signal
ylabel('Filtered Speech Signal');
xlabel('Sample Number');
for i=100:1100%Storing 1001 sample values
    x2(i)=y.pcm(i);
end
[x2_DTFT,W2]=DTFT(x2,0);
plot(W2,abs(x2_DTFT));%Plotting DTFT of the Filtered signal
ylabel('Magnitude(Filtered Response)');
xlabel('Frequency in rad/sec');
plot(W2,abs(x2_DTFT));%Plotting DTFT of the Filtered signal in a cetain range
around w
xlim([w-0.02,w+0.02]);
ylabel('Magnitude(Filtered Response)');
xlabel('Frequency in rad/sec');
```

e. No, we cannot use r=0.999999 to get the filtered signal. We can see below the time domain plot of the filtered signal:

```
x=IIRfilter(y.pcm,w,0.999999);
sound(x);
plot(x);%Plot of the filtered signal
ylabel('Filtered Speech Signal R=0.999999');
xlabel('Sample Number');
```



Here the reason for this alteration is over-filtering. Due to increasing r, we got a very sharp filter which is cutting off the noise signal as well as the actual signal frequencies so getting the distorted time domain signal.

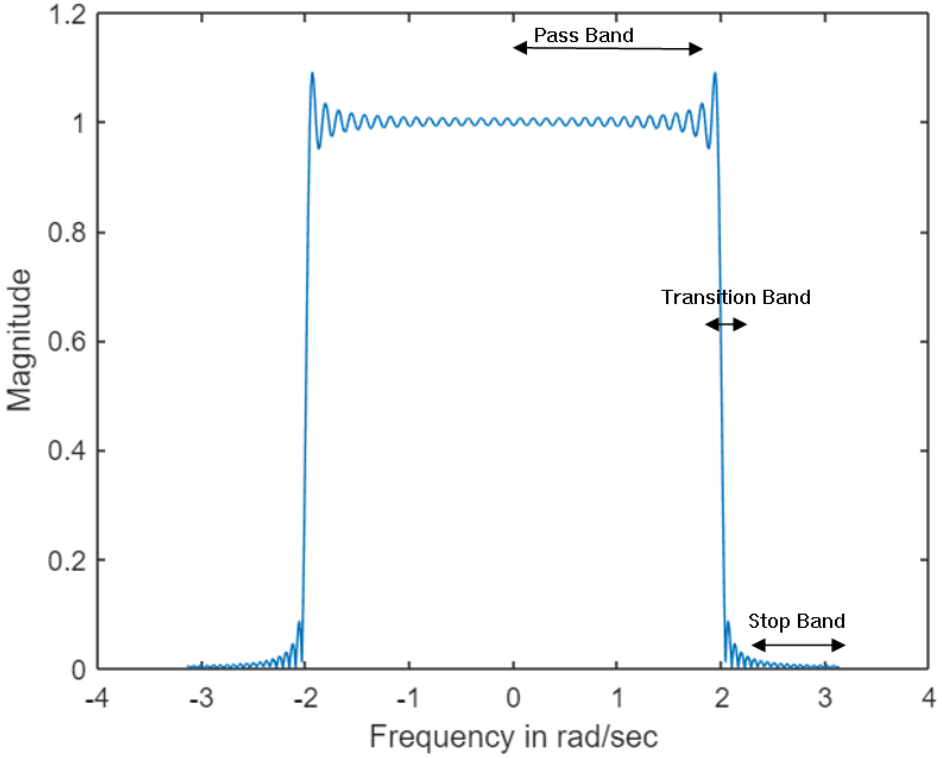
5.

Lab Report:

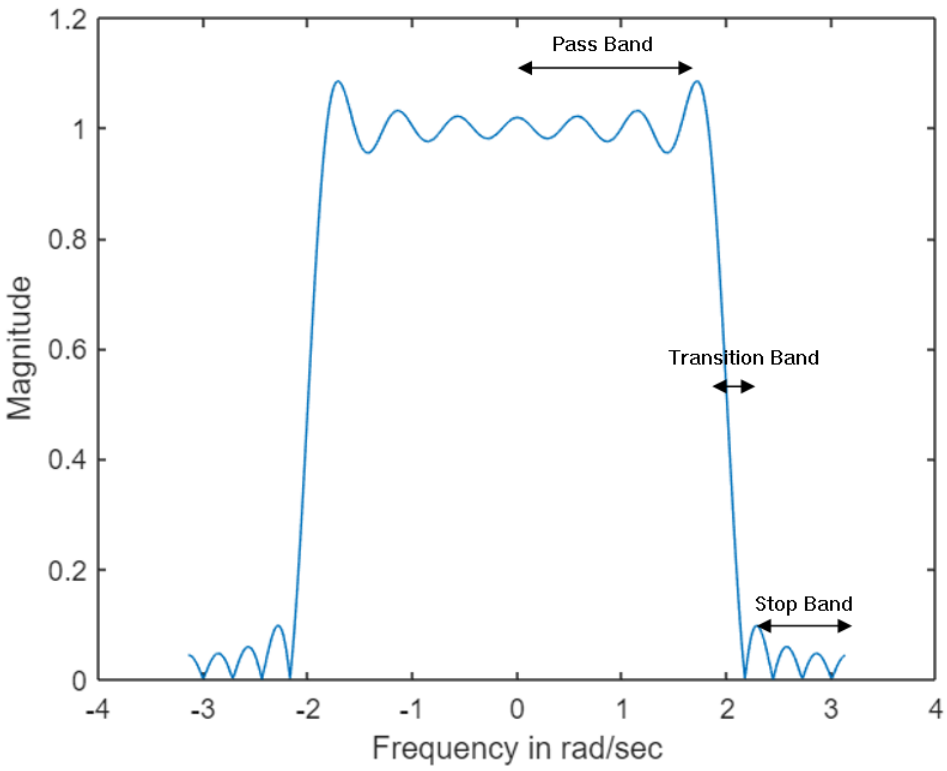
- Submit the plots of the magnitude response for the two filters (not in decibels). On each of the plots, mark the passband, the transition band and the stopband.
- Submit the plots of the magnitude response in decibels for the two filters.
- Explain how the filter size effects the stopband ripple. Why does it have this effect?
- Comment on the quality of the filtered signals. Does the filter size have a noticeable effect on the audio quality?

a. Plots of the Magnitude response of the two filters, i.e., for N=101 & N=21:

i. N=101:

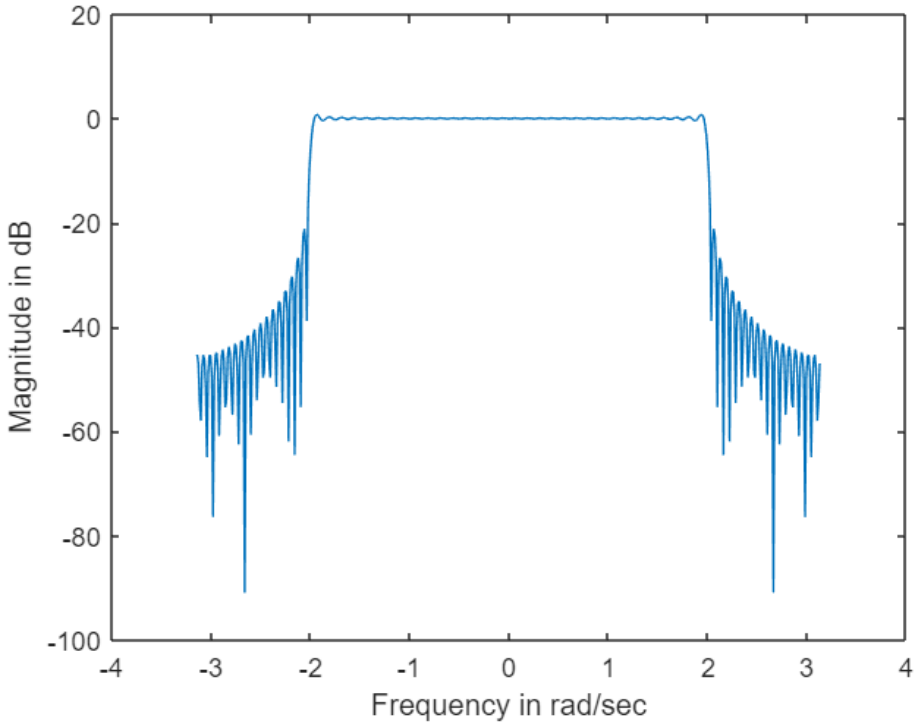


ii. N=21:

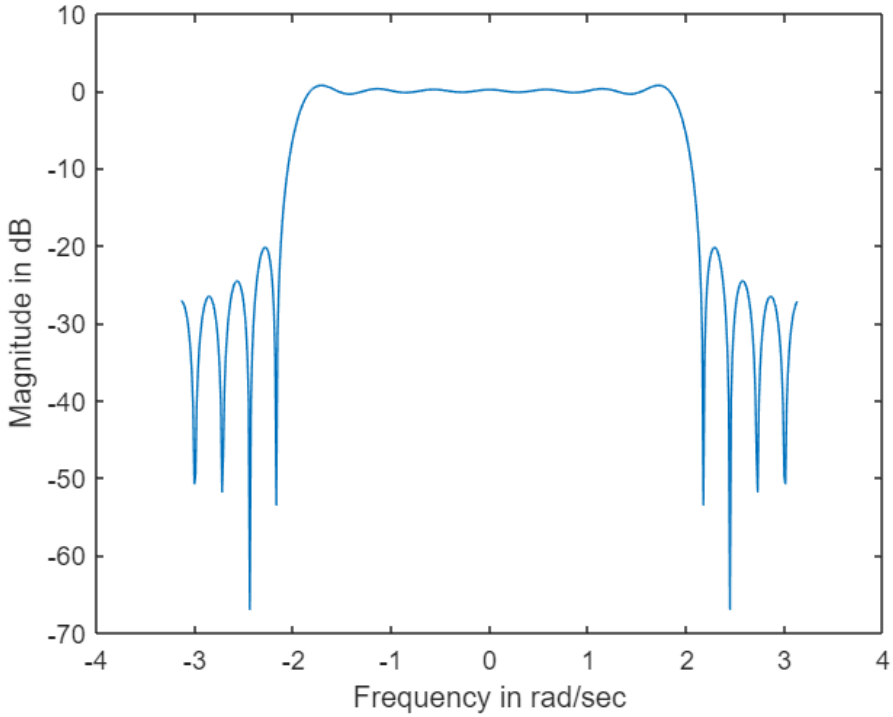


b. Plots of the Magnitude response in decibels of the two filters, i.e., for N=101 & N=21:

i. N=101:



ii. N=21:



c. Explanation:

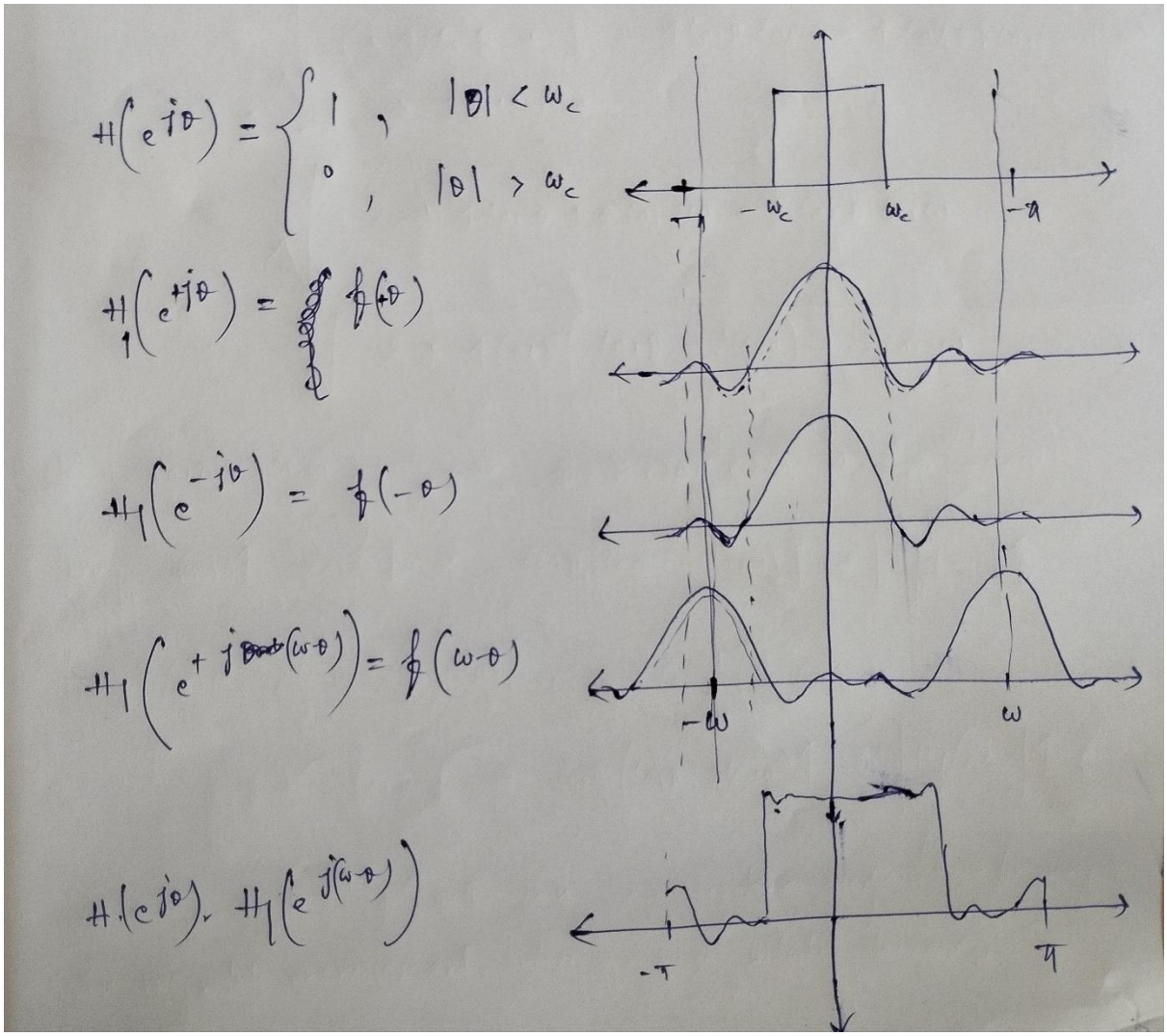
$$h[n] = \frac{\omega_c}{\pi} \operatorname{sinc}\left(\frac{\omega_c n}{\pi}\right), \quad -\infty < n < +\infty$$

$$h_1[n] = \begin{cases} 1, & |n| \le M \\ 0, & |n| > M \end{cases}$$

So, $$h[n] \cdot h_1[n] = h_{trunc}[n] = \begin{cases} \frac{\omega_c}{\pi} \operatorname{sinc}\left(\frac{\omega_c n}{\pi}\right), & |n| < M \\ 0, & |n| > M \end{cases}$$

So, $$H_{trunc}(e^{j\omega}) = DTFT\left(h[n] \cdot h_1[n]\right)$$

$$= \int_{2\pi} H(e^{j\theta}) \cdot H_1(e^{j(\omega-\theta)}) \cdot d\theta$$



From the last waveform, we can see that some ripple came in the Stopband as well as the pass band. Now doing the integration will introduce more ripple in it.

d. Comment:

For the more sample filter, we got a clearer sound than the less sample filter.

e. MATLAB CODE and extra waveforms:

i. Overall code:

```
y=load("C:\Users\PANTHADIP MAJI\Desktop\DSP_LAB\Exp_5\nspeech2.mat");
sound(y.nspeech2);
plot(y.nspeech2);%Plot of the unfiltered signal
ylabel('Unfiltered Speech Signal');
xlabel('Sample Number');


h=LPFtrunc(101);%Generating 101 lengthed impulse response of the filter
[DTFT_h,W]=DTFT(h,512);
```

```
plot(W,abs(DTFT_h));%Plotting DTFT of the impulse response of the filter
ylabel('Magnitude');
xlabel('Frequency in rad/sec');
plot(W,20*log10(abs(DTFT_h)));%Plotting DTFT of the impulse response of the
filter in dB
ylabel('Magnitude in dB');
xlabel('Frequency in rad/sec');
filtered_sig=truncFilter(h,y.nspeech2);
plot(filtered_sig);%Plot of filtered signal
ylabel('Filtered Speech Signal');
xlabel('Sample Number');
sound(filtered_sig);


h=LPFtrunc(21);%Generating 101 lengthed impulse response of the filter
[DTFT_h,W]=DTFT(h,512);%Plotting DTFT of the impulse response of the filter
plot(W,abs(DTFT_h));
ylabel('Magnitude');
xlabel('Frequency in rad/sec');
plot(W,20*log10(abs(DTFT_h)));%Plotting DTFT of the impulse response of the
filter in dB
ylabel('Magnitude in dB');
xlabel('Frequency in rad/sec');
filtered_sig=truncFilter(h,y.nspeech2);
plot(filtered_sig);%Plot of filtered signal
ylabel('Filtered Speech Signal');
xlabel('Sample Number');
sound(filtered_sig);
```

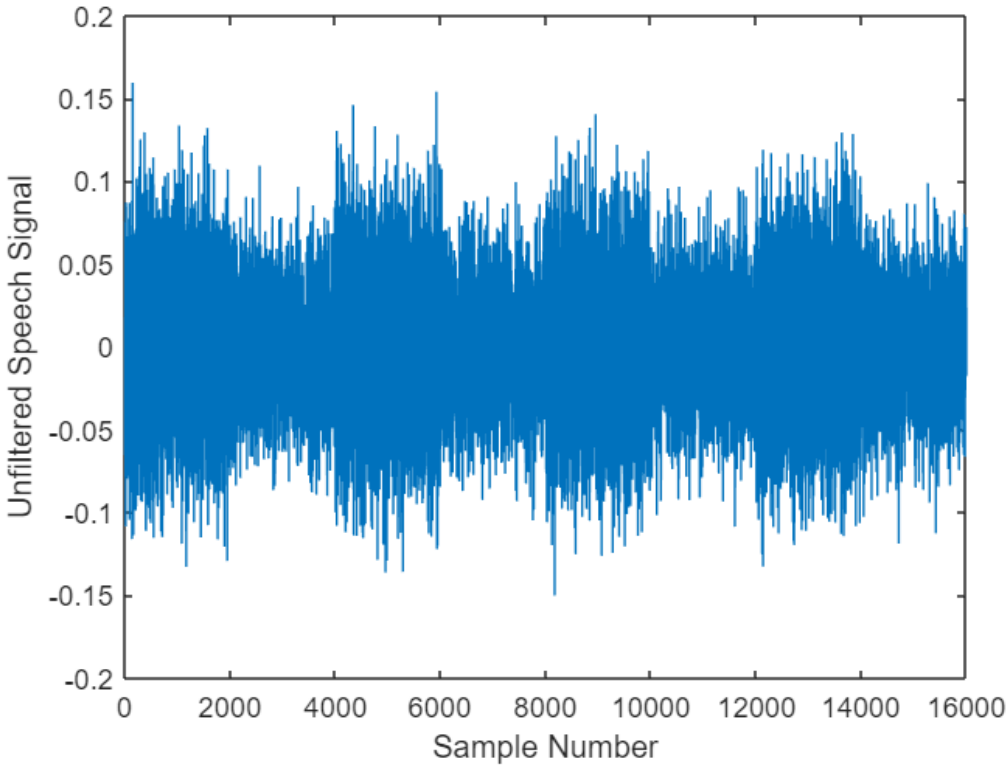
ii. Code for truncFilter function:

```
function x=truncFilter(sig,filter)
x=conv(sig,filter);
end
```

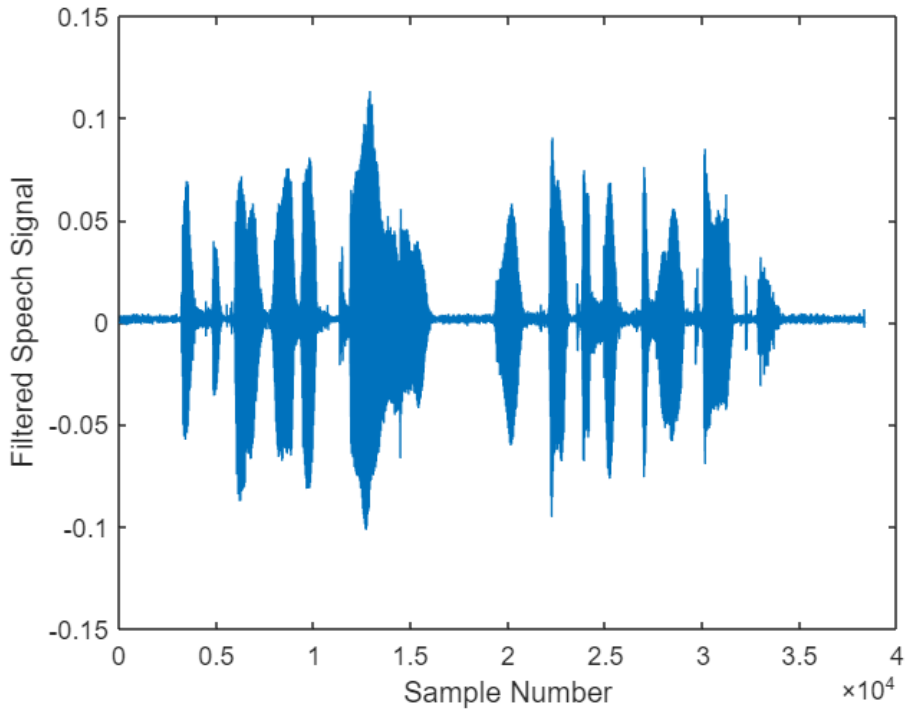
iii. Code for impulse response creating function of the filter:

```
function h=LPFtrunc(N)
j=(-1)^0.5;
x=zeros(N,1);
for i=0:N-1
    x(i+1)=(2/pi)*sinc(2*(i-((N-1)/2))/pi);
end
h=x;
end
```

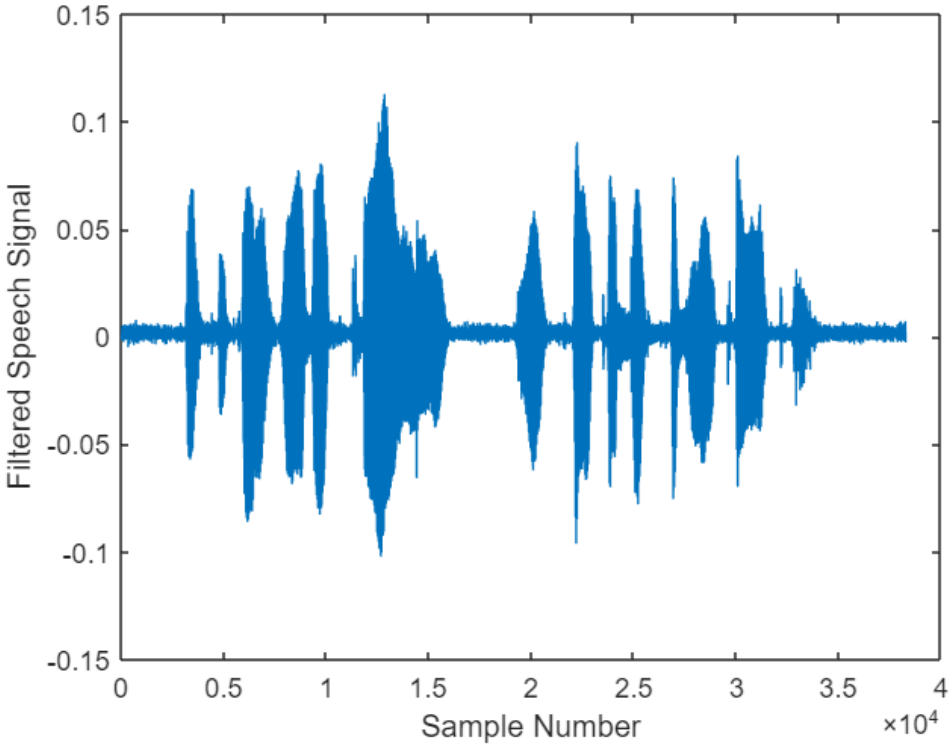
i. Unfiltered waveform:



ii. Filtered waveform when N=101:



iii. Filtered waveform when N=21:



f. The Dominant Frequency(w) of the unfiltered signal:

```
x1_max = 26.6547
pos_max = 1750
w = -2.4709
```

Digital Signal Processing Laboratory (EE39203)

Experiment 2:

Discrete Time Systems

Slot: Friday, Name: Panthadip Maji, Roll No: 20EE30019

Extra Assignment:Question: Create a sinusoidal signal. Add noise to it. Then run it through a moving average function of different window sizes. Comment on the result.

Ans:

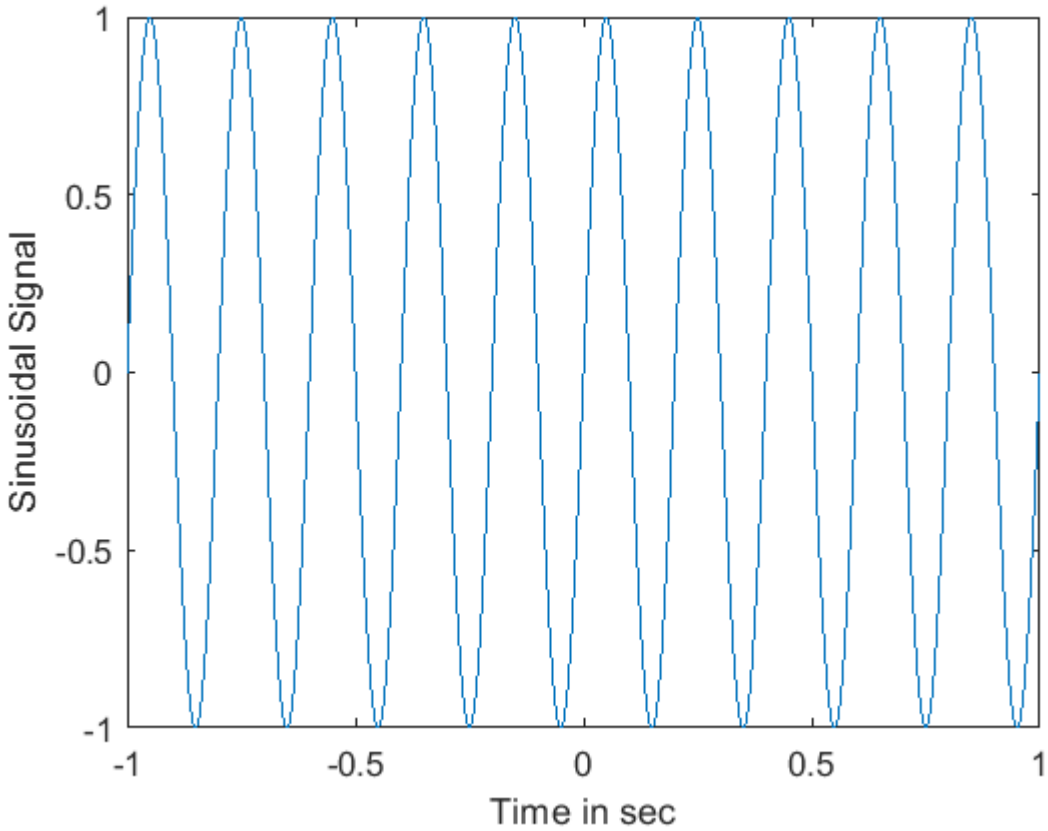
Matlab Code:

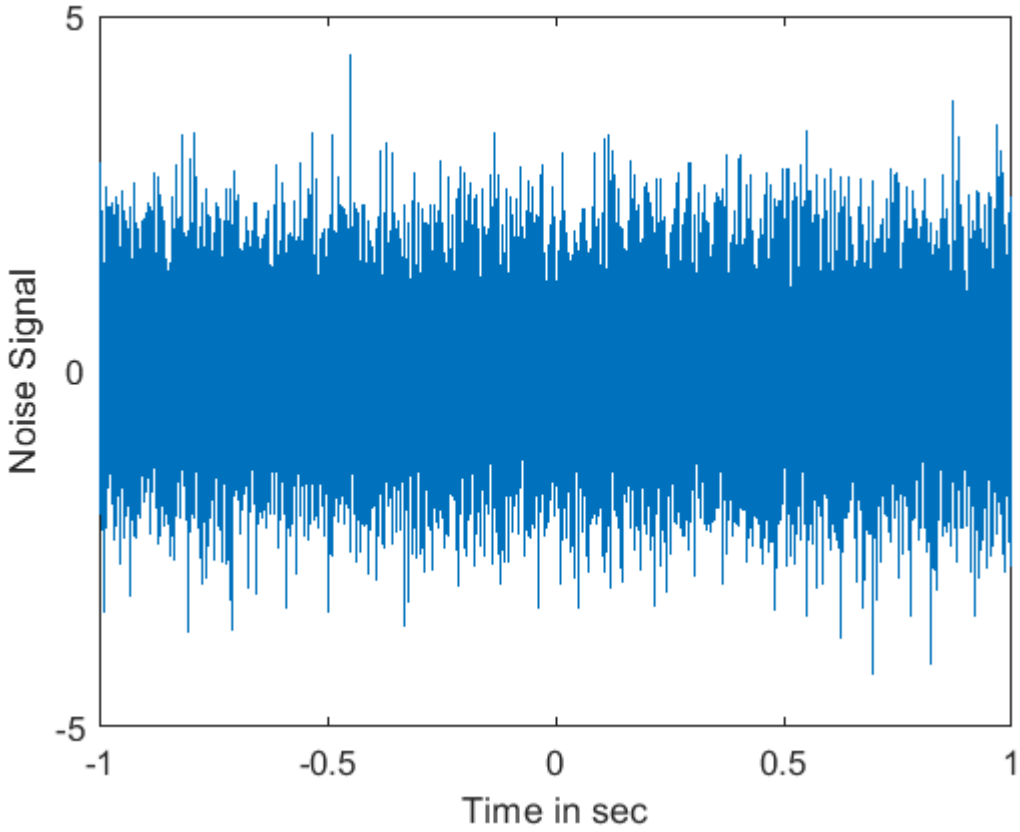
```
%Creating A random function
f=5;%Frequency in Hz of the sinusoidal signal
fs=10000;%Sampling rate of the signal
N=2;%Time Duration of the signal
t=-N/2:1/fs:N/2;
x=zeros(N*fs+1,1);
z=zeros(N*fs+1,1);
y=sin(2*pi*f*t);
plot(t,y);
ylabel('Sinusoidal Signal');
xlabel('Time in sec');


%Creating the noise signal
for i=1:N*fs+1
    x(i)=randn;
end
plot(t,x);
ylabel('Noise Signal');
xlabel('Time in sec');


%Adding noise & the sinusoidal signal
for i=1:N*fs+1
    z(i)=x(i)+y(i);
end
plot(t,z);
ylabel('Noise+sinusoid Signal');
xlabel('Time in sec');


%Moving Average implementation with different window sizes
w=Moving_Average(z,11);%Window size=11
subplot(3,1,1);
plot(t,w);
ylabel('Signal');
xlabel('Time in sec');
title('After implementation of moving average of size 11');
w=Moving_Average(z,101);%Window size=101
subplot(3,1,2);
plot(t,w);
ylabel('Signal');
xlabel('Time in sec');
title('After implementation of moving average of size 101');
w=Moving_Average(z,1001);%Window size=1001
subplot(3,1,3);
plot(t,w);
ylabel('Signal');
xlabel('Time in sec');
title('After implementation of moving average of size 1001');
```

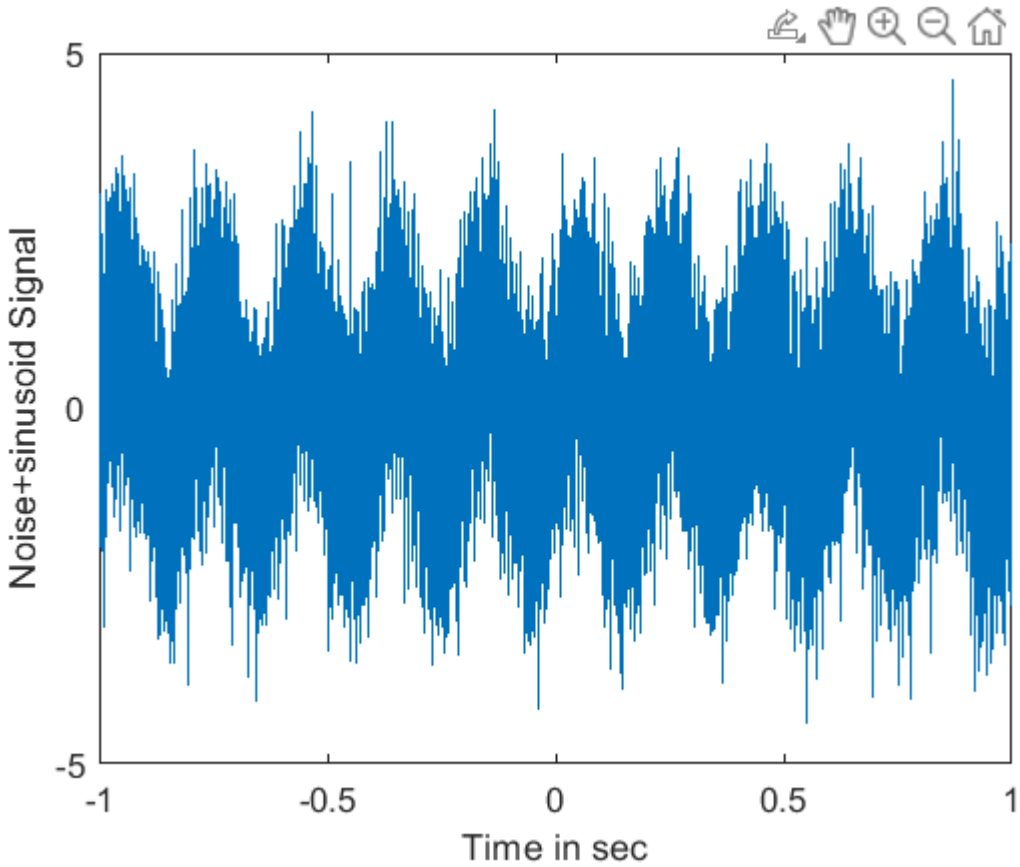
Corresponding Moving Average function:

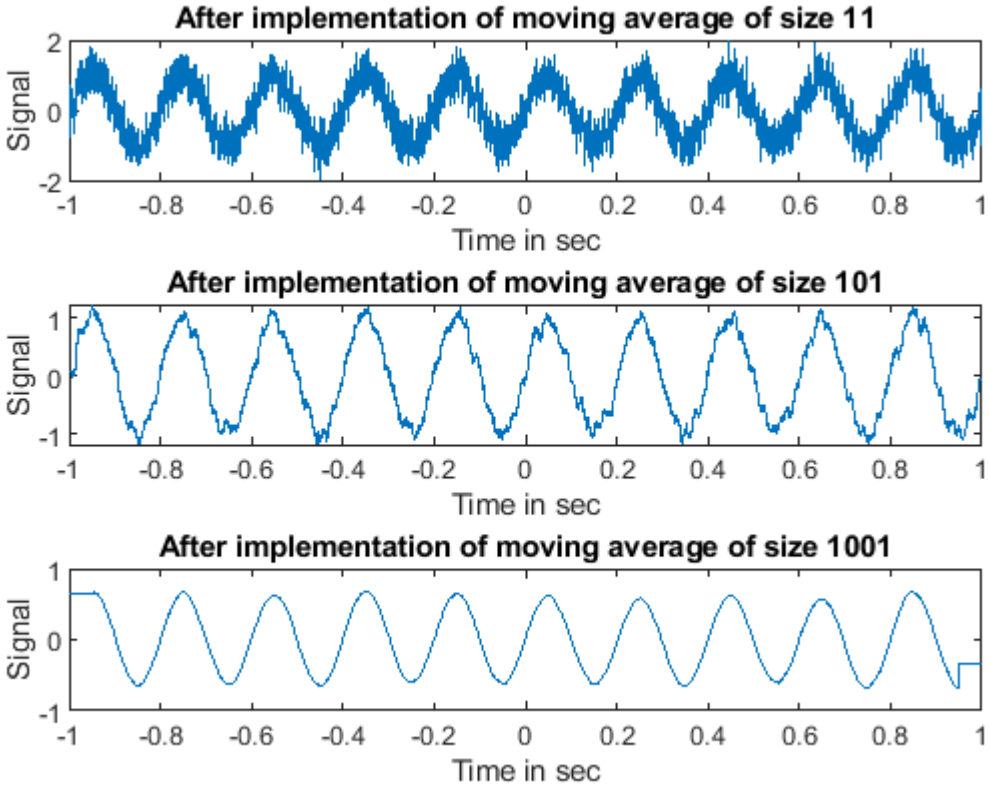
```
function x=Moving_Average(z,K)
N=length(z);
dummy=zeros(N,1);
for i=1:N
if i<=(K+1)/2
    for j=1:K
        dummy(i)=dummy(i)+z(j)/K;
    end
end
if i>(K+1)/2 && i<(N-(K+1)/2)
     for j=0:K-1
         dummy(i)=dummy(i)+z(i-((K-1)/2)+j)/K;
     end
end
if i>=N-(K+1)/2
     for j=N-(K+1)/2:N
        dummy(i)=dummy(i)+z(j)/K;
     end
end
end
```

Corresponding Plots:




5
0
-5
Noise Signal
-1
-0.5
0
0.5
1
Time in sec


5
0
-5
Noise+sinusoid Signal
-1
-0.5
0
0.5
1
Time in sec



Comment: Here we can see that if the window size increases then we get a smooth and more errorless signal.